

軟體測試

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授課時間：每週三 9：10—12：00

博理館 103

課程代號：EE 5170

識別碼：921 U9550

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期末考

考試中，請勿參閱資料。

1. In **Figure 1** is a procedure in Python.

```
def searchPattern(a, b, fn):
    if type(a) != int or type(b) != int or type(fn) != str:
        return None
    if not os.path.isfile(fn):
        return None
    with open(fn, 'r') as f:
        s = f.read()
    count = 0
    for k in range(len(s)):
        if s[k:(k+a)] in s[k:(k+b)] or s[k:(k+b)] in s[k:(k+a)]:
            count += 1
    return count
```

Figure 1: a simple Python method

Please draw the control-flow graph of the above program. (5pts/5)

2. Continuing from question 1, please list the test requirements for prime-path coverage (PPC). (5pts/10)

3. Continuing from question 2, please write down a test set for the full PPC. (5pts/15)

4. Continuing from question 1, please list all of the du-paths with respect to s , $count$, and k . (Note: Include all du-paths, even those that are subpaths of some other du-path). (5pts/20)

5. Continuing from question 4, please write down a minimal test set that satisfies all-defs coverage with respect to s , $count$, and k . (Direct tours only.) (5pts/25)

6. Continuing from question 5, please write down a minimal test set that satisfies all-uses coverage with respect to s , $count$, and k . (Direct tours only.) (5pts/30)

7. Suppose we have the following predicate:

$\text{type}(a) \neq \text{int} \text{ and } (\text{type}(b) \neq \text{int} \text{ or } (a \geq b \text{ and } \text{type}(fn) == \text{str}))$

Please identify the clauses (*atoms* in standard logic terms) that determine this predicate.

(5pts/35)

8. Continuing from question 7, please compute (and simplify) and list the conditions under which each of the clauses determines the predicate. (5pts/40)

9. Continuing from question 8, write the complete truth table for all clauses. Label your rows starting from 0 at the top. Use the format that row 0 should be all clauses true. You should include columns for the conditions under which each clause determines the predicate, and also a column for the predicate itself. (5pts/45)

10. Continuing from question 9, identify all pairs of rows from your table that satisfy *general active clause coverage (GACC)* with respect to each clause. (5pts/50)

11. Continuing from question 9, identify all pairs of rows from your table that satisfy *correlated active clause coverage (CACCC)* with respect to each clause. (5pts/55)

12. Please explain the STRIDE categories of threat models in security testing. (5pts/60)

13. Please consider black-box testing to the method in Figure 1. There is no other document. There are four inputs, including, a, b, fn, and the contents of the file with name in fn, to the method. Please define characteristics of inputs to this method and list them. (5pts/65)

14. Continuing from question 13, please partition the characteristics into blocks and write them down. (5pts/70)

15. Continuing from question 14, please define and list the test values for the blocks.
(5pts/75)

16. Assume that we have the procedure in **Figure 1** to test. Please write down a mutant with the Arithmetic Operator Replacement (AOR) and write down a test case (values for a and b) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (5pts/80)

17. Please define and explain a coverage metric for testing convolutional neural network (CNN). (Ref: *DeepTest: Automated Testing of Deep-Neural-Network-Driven Autonomous Cars*, ICSE 2018; *Towards Improved Testing For Deep Learning*, ICSE : New Ideas and Emerging Results (ICSE-NIER) 2019) (5pts/85)

18. Please explain how to use file difference tools to reduce the cost of mutation coverage analysis in regression testing. (Ref: *State of Mutation Testing at Google*, ICSE 2018) (5pts/90)

19. Please explain the difference between implementation bugs and model bugs in applications that use machine learning technology. Please also explain how to test for such implementation bugs. (Ref: *Identifying Implementation Bugs in Machine Learning Based Image Classifiers using Metamorphic Testing*, ISSTA 2018) (5points/95)

20. Please explain the difference between the sequential consistency (SC) memory model and the total store order (TSO) memory model. Please also give a (hypothetical) example of thread-level bugs that TSO memory model may incur. (Ref: *Thread-Modular Static Analysis for Relaxed Memory Models*, ESEC/FSE 2017) (5points/100)